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Bidirectional Gated Recurrent Unit with Glove Embedding and Attention Mechanism for Movie Review Classification

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Abstract

During the current Internet era, enormous volume of structured and unstructured textual data is generated and exchanged online that made the text classification more crucial. With the massive amount of user-generated content on social media platforms, businesses and organizations can leverage sentiment analysis to understand customer opinions and attitudes towards the products, services, or brand. Text classification is a natural language processing task in which a machine learning model is trained to categorize the text into predefined classes or categories. Due to the advent of promising Deep Learning techniques,intelligent and accurate text classification system can be developed. In movie review classification, algorithms are used to automatically categorize movie reviews into positive, negative, or sometimes neutral sentiments based on the opinions expressed in the text. The goal is to automate the process of determining whether a review reflects a positive or negative sentiment, helping users quickly understand the overall reception of a movie. This work proposes an automated movie review classification system based on Bidirectional Gated Recurrent Unit encoder with attention module. This method encodes the words using GloVe word embedding technique. Context information is captured in an efficient manner and the encoding is more reliable. Bidirectional GRU layer captures long-term dependencies in sentences, followed by a custom Attention layer that assigns higher weights to key components. The attention mechanism allows the model to selectively focus on important parts of the input sequence, by dynamically determining attention weights based on task relevance.The model was individually trained and evaluated using popular benchmark IMDb dataset. Experimental studies have proved the better performance of proposed movie review classification system that obtained accuracy of 98%.

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Keywords: Attention mechanism; Bidirectional Gated Recurrent Unit; GloVe embedding; Text classification; Sentiment analysis.

1. Introduction

The amount of complicated documents and texts has grown exponentially in recent years. Natural Language Processing (NLP) plays a pivotal role contributing to the effective analysis of textual data for sentiment classification. In the preprocessing phase, NLP techniques are employed to clean and transform raw text into a format suitable for model training. The removal of punctuation, lemmatization, and lowercasing, facilitated by NLTK, ensures that the model operates on standardized and semantically meaningful representations of words. Text classification[1] as-

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signs a list of predetermined categories to open-ended text. Text classifiers can be used to organize, arrange, and categorize any type of text, including files from the web, medical research, publications and other data available in internet.The text classification applications include sentiment analysis, email filtering, document categorization[2], news classification[3], medical diagnosis[4], legal document analysis, fraud detection, movie review classification[5], language identification, social media analysis, and customer support applications. Text classification makes unstructured text data more accessible and valuable, benefiting various industries and improving decision-making and user experiences.

Text classification offers numerous advantages in our day-to-day lives. It is widely used in email systems to filter and organize incoming messages, in news curation to categorize articles into relevant topics, and in customer support to efficiently address queries and direct users to the appropriate channels. Sentiment analysis, powered by text classification, enables businesses to understand customer opinions and emotions, while spam detection ensures users are protected from harmful content. Personal assistants include Siri and Alexa rely on text classification to interpret user commands accurately. Additionally, recommendation systems use it to understand user preferences and offer personalized content. In healthcare, text classification aids medical research and diagnosis by analyzing patient records and research articles. Finance utilizes text classification for sentiment analysis to inform investment decisions. Language translation also benefits from text classification to accurately determine the source language, facilitating more precise translations. These applications showcase the pervasive impact and indispensable role of text classification in enhancing our daily interactions with technology and information.The reason we want to classify movie reviews is to understand what people think about movies without reading every single review. This helps businesses shape their strategies and make things better for users. But, it's tricky because reviews can be tricky, with sarcasm and different ways of saying things. Also, keeping up with how people express themselves about movies adds another layer of difficulty to building models that get it right.

The research objective is to develop a method for efficient movie review sentiment classification based on deep learning techniques and attention mechanism. The contribution of this work is two fold. This paper explores the possibility of utilizing the GloVe embedding for movie review classification. The bidirectional Gated Recurrent Unit (GRU) captures the sentiments from the movie review and the model captures the sentiments and classify as positive or negative reviews.

2. Review of Literature

Author et al.	**Authors' work**	**Model used**
Zheng, J. et al(2019)[18]	Bidirectional RNN and Attention-Based Model for Text Classification	BRCAN
Liu, C.,et al(2022)[14]	Text Classification with Document-Relational Graph CNN	Graph Convolutional Networks
Li, Q.,et al.(2022)[13]	Review on text classification	Machine Learning and Deep Neural Networks
Xu, Y., et al.(2022)[15]	Adaptive Dense Ensemble Model for Text Classification	AdaDEM and EnCNN
Gong, J.,et al(2020)[16]	Hierarchical graph transformer-based deep learning model for large-scale multi-label text classification	Hierarchical graph transformer-based model

Table 1. Literature Review

3. Proposed Method- Bidirectional Gated Recurrent Unit With Glove Embedding and Attention Mechanism for Movie Review Classification

The proposed method is categorized into four modules. Word embedding module, Bidirectional GRU module for capturing context information, Attention module and Classification module This figure represents the architecture of the model with that illustrates word embedding and hybrid neural network. First the preprocessed data is passed to the input layer. In the embedding layer, GloVe embedding model is used for the representation of words into vectors.The GloVe layer utilizes pre-trained word embeddings, such as those from GloVe, to provide meaningful numerical representations of words based on their semantic contexts. In the context of movie review classification, the GloVe layer proves advantageous as it captures nuanced language nuances and sentiment, improving the model's ability to understand and classify reviews based on the rich semantic information embedded in the pre-trained word vectors. The long-term dependency between the words in the sentences are given to Bi-directional GRU layer to learn. The intermediate representations generated by Bidirectional GRU are given to the dense layer. Next the attention layer is used to give them higher weights to capture the key components in texts.

The more the data is clean, better the solution is. The dataset is taken from IMDb dataset. The unnecessary columns are dropped to retain only relevant information. After that to handle Sentiment Labels, the values are converted into binary values. Text Cleaning can be done by converting all text to lowercase to make it case-insensitive, Lemmatizing each word in the movie review to reduce words to their base or root form and by removing stopwords, the noise in the data are thus reduced by text cleaning.

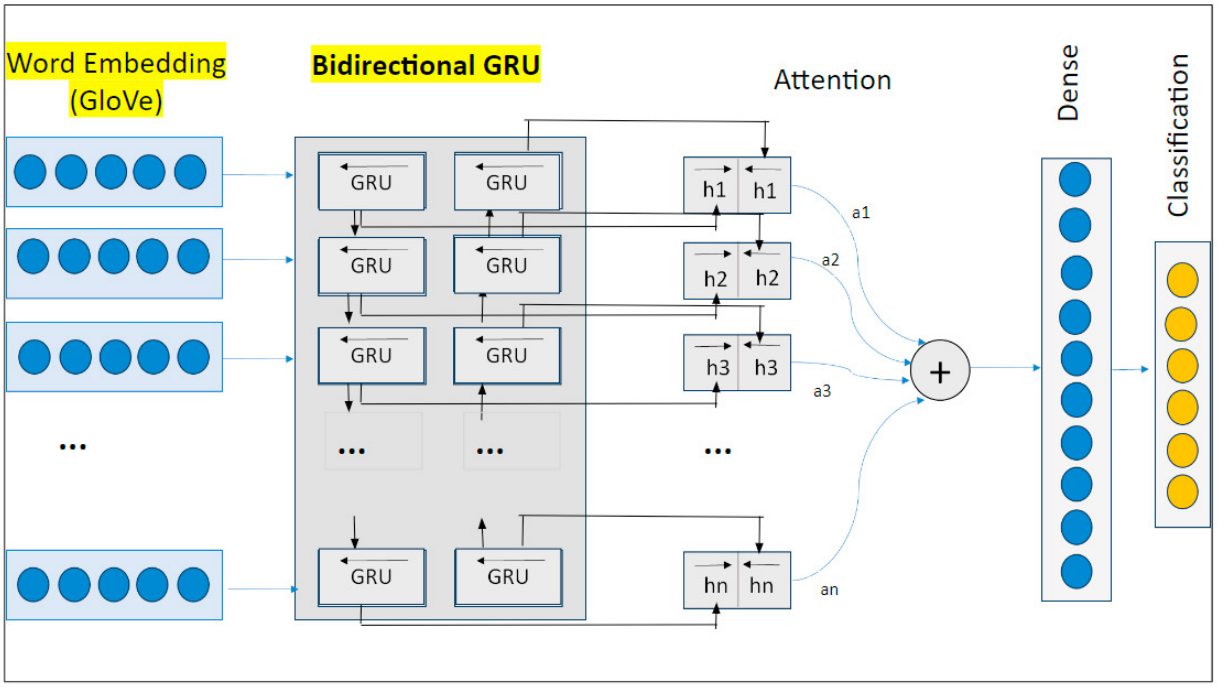


Fig. 1. Block diagram of the proposed methods Bidirectional GRU with GloVe Embedding and Attention mechanism for classifying movie reviews

The text preprocessing[6] and data preparation process for building a text classification model uses the tokenizer class to tokenize the text data. It begins by building a vocabulary, assigning a unique integer index to each word based on its frequency in the provided text corpus. Tokenizer can pad sequences to a specified length, ensuring uniformity for model input. It also offers mechanisms for handling out-of-vocabulary words, allowing for flexibility in dealing with unseen terms. The model architecture includes an embedding layer with word embeddings of size 128 and an Recurrent Neural Network layer with 32 units. The maximum sequence length for the text data is set to 130, and all sequences are padded with zeros or truncated to this length. The target labels for the movie sentiment analysis

are extracted. This preprocessing prepares the input data for a classification model, where the sequences are of equal length, and the sentiment labels are ready for training. The subsequent step would involve building and training a text classification model using the prepared data to predict the sentiment of new movie review data.

The attention mechanism[7] enables the model to selectively focus on important parts of the movie reviews during prediction. The attention weights are learned during the training process. The neural network learns to assign higher weights to more relevant words or tokens, and these weights are used to compute a weighted sum, which forms the context vector. The attention weights are essentially a measure of the importance of each word or token in influencing the model's prediction. The learning process involves optimizing these weights to improve the overall performance of the movie sentiment analysis task, emphasizing words that contribute more to the sentiment expression in movie reviews.The Attention class uses three dense layers to learn attention weights, that determine the significance of words in the input sequence. Then performs calculations to obtain attention scores and attention weights using a combination of tanh activation and softmax function. The resulting attention weights are then used to compute a context vector that captures the most relevant information from the input sequence. This custom attention layer can significantly improve the performance of model, particularly when handling long sequences or tasks involving complex dependencies among sequence elements. To effectively manage long sequences in the model, various strategies are employed. One approach involves truncating lengthy sequences to a fixed length and padding shorter ones to match the length of the longest sequence, though the latter may lead to information loss. In this work, attention technique is utilized to handle the long sequences. Attention mechanisms offer another powerful solution, enabling the model to selectively focus on different segments of the sequence, assigning higher importance to relevant portions. For exceptionally long sequences, segmentation can be employed, breaking the sequence into smaller parts processed independently. The model architecture includes recurrent layers, they can implicitly handle sequential data, potentially incorporating a form of segmentation.

After that loads pre-trained GloVe word embeddings[8] and creates an embedding matrix that can be used as initial weights for the embedding layer of a movie review classification model. The embedding matrix is based on the words present in the tokenizer vocabulary, ensuring that only the most frequent words are considered and the embedding layer is initialized with meaningful word representations. This technique helps in improving the model performance by leveraging pre-trained word embeddings and transferring knowledge from a large corpus to the specific text classification task at hand. The model architecture consists of an embedding layer with pre-trained word embeddings, a Bidirectional GRU layer to capture bidirectional context, a custom Attention layer to focus on significant parts of the review texts, and softmax for classification.

3.1. GloVe based Word Embedding

GloVe (Global Vectors for Word Representation) is a popular word embedding technique in Natural Language Processing(NLP). It constructs word vectors by analyzing the global co-occurrence statistics of words in a large text corpus. GloVe's objective function minimizes the difference between dot products of word vectors and the logarithm of their co-occurrence probabilities, resulting in dense vector representations where semantically related words are closer together. Pre-trained GloVe embeddings are available for multiple languages and have proven valuable in NLP tasks such as sentiment analysis and machine translation. While GloVe captures global word relationships effectively, it may require substantial computational resources for training and may not capture fine-grained semantic nuances.

3.2. Bidirectional Gated Recurrent Unit

A Bidirectional GRU, or BiGRU[9], is a sequence processing model that consists of two GRUs. one taking the input in a forward direction, and the other in a backwards direction. It is a bidirectional recurrent neural network with only the input and forget gates. The input sequence, typically composed of word embeddings, is processed in two passes: a forward pass from the beginning to the end and a backward pass from the end to the beginning. During the forward pass, the forward GRU computes the hidden state at each time step based on the current input word and the previous hidden state. The reset gate and update gate mechanisms allow the model to determine what information to forget and what to retain. Similarly, during the backward pass, the backward GRU computes the hidden state at each time step based on the current input word and the previous hidden state. After both passes, the hidden states from

the forward and backward GRUs are concatenated to create a final hidden representation for each word in the input sequence. This bidirectional processing enables the BiGRU to capture contextual information from both the past and future contexts of each word, making it highly effective in tasks that require understanding of the overall sequence context. The concatenated hidden states are then used as the final representation of the input sequence. Depending on the specific task, additional layers like fully connected or softmax layers can be added for tasks like sentiment analysis, machine translation, or named entity recognition. BiGRU has demonstrated excellent performance in natural language processing tasks and is widely used in deep learning based NLP models.

3.3. Attention Module

The attention mechanism is a crucial concept in deep learning, particularly in natural language processing and computer vision. It allows neural networks to selectively focus on significant attributes or features of the input data, improving their performance by assigning weights to different elements based on their importance. This attention mechanism, commonly seen in Transformer models, enables capturing long-range dependencies and contextual understanding of sequences in NLP tasks. It also helps computer vision models focus on essential regions in images, enhancing their ability to recognize complex patterns. Overall, the attention mechanism significantly enhances the capabilities of deep neural networks in various applications.

While using attention mechanism in classifying text, it uses a sophisticated NLP technique that concentrates on significant words or phrases. The input text is transformed into numerical representations, and attention weights are dynamically determined based on task relevance. RNNs, LSTMs, and Transformers are examples of typical model that incorporate attention to gather contextual data.

The proposed work attempts to build an efficient model for movie review analysis that uses attention mechanism to obtain an accurate result.GloVe, short for Global Vectors for Word Representation, is an unsupervised learning algorithm used to create word embeddings in natural language processing (NLP). Word embeddings are dense vector representations of words in a continuous space, where semantically similar words are represented by vectors that are closer together. Bidirectional Gated Recurrent Unit, a neural network architecture used in natural language processing tasks. It combines bidirectional processing, capturing information from past and future time steps, with Gated Recurrent Units (GRUs), that use gating mechanisms to control information flow and address the vanishing gradient problem. This combination enables BiGRU models to effectively capture contextual information and long-range dependencies in input sequences for language modelling, machine translation and sentiment analysis. Based on the context vector and the hidden state of the decoder, the attention weight for each time step is evaluated based on the computation of score calculation, attention weights. The context vector can be computed as the weighted sum of the input sequence.

3.4. Softmax Classifier

The softmax function plays a key role classifying the moview reviews. The attention mechanism calculates scores for each time step in the input sequence, determining their relevance to the current hidden state. After applying the hyperbolic tangent activation to these scores, the softmax classifier computes a probabilities for each movie whether the comments are positive or negative.

For movie review classification using softmax classifier, the process involves converting text data into a numerical format and designing a neural network with an embedding layer. The model is trained to predict sentiment categories such as positive, negative and neutral through softmax function in the output layer. This classifier transforms the raw output into a probability distribution, allowing the model to assign probabilities to each sentiment class. The model is trained using categorical crossentropy loss and optimized with an algorithm like stochastic gradient descent. The result is a trained model capable of classifying movie reviews into different sentiment categories based on learned patterns in the input data.

3.5. Hyperparameters used

In sentiment analysis, several hyperparameters are employed to configure and optimize the deep learning model. These hyperparameters include MAX_FEATURES (limiting vocabulary size), EMBED_SIZE (word embedding dimensionality), RNN_CELL_SIZE (number of units in RNN layers), MAX_LEN (maximum input sequence length), and Batch Size (BATCH_SIZE) and Number of Epochs (EPOCHS) for training control. DROPOUT RATE is fixed as 0.05 for neural network architecture and regularization. These hyperparameters play critical roles in shaping the model performance and generalization. Fine-tuning these hyperparameters is essential, involving experimentation to discover the most suitable values for a sentiment analysis task.

Hyperparameters	Value
MAX_FEATURES	6000
EMBED_SIZE	128
RNN_CELL_SIZE	32
MAX_LEN	130
BATCH_SIZE	100
EPOCHS	5
DROPOUT	0.05

Table 2. Hyperparameters used

4. Experimental Results

To assess the general performance of our sentiment analysis model for movie reviews, we employed a test-train split on the dataset. The dataset, comprising a diverse collection of movie reviews with corresponding sentiment labels, was divided into two subsets: a training set and a testing set. The split was stratified to ensure that the distribution of sentiment classes remained consistent between the training and testing sets. Specifically, 80% of the data was allocated to the training set, used for training the model, while the remaining 20% constituted the testing set. Then our proposed system which includes Bidirectional Gated recurrent unit with glove embedding and attention is tested against a dataset which gives an accuracy of 98%.

Accuracy	98%
Precision	98%
Recall	98%

Table 3. Experimental Results of the Work

4.1. Dataset Description

The proposed approach was trained and tested separately on the train and test splits of the IMDb dataset.The IMDb dataset is a publicly available dataset widely used in natural language processing and sentiment analysis tasks. It consists of movie reviews along with their associated sentiment labels (positive or negative).The dataset contains 25000 positive labelled reviews and 25000 negatively labelled reviews. The dataset is sourced from the IMDb website, a popular movie database.

4.2. Comparison with other models

Various models have been explored for sentiment analysis on IMDb datasets, showcasing diverse approaches to achieve high accuracy. The XLNet model achieved an accuracy of 96.21%, while a combination of Bidirectional Long Short-Term Memory (BiLSTM) with attention and Convolutional Neural Network (CNN) yielded a

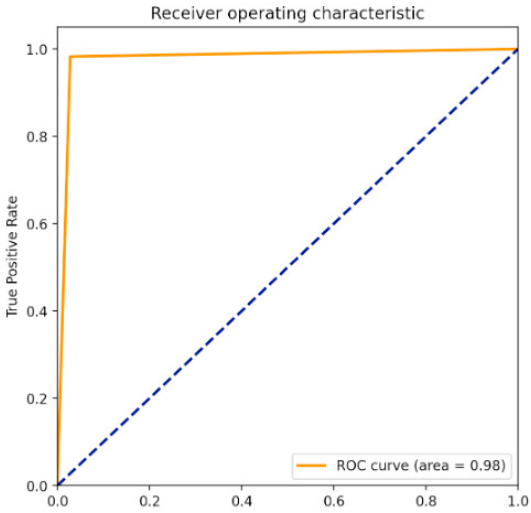


Fig. 2. ROC Curve of Proposed System

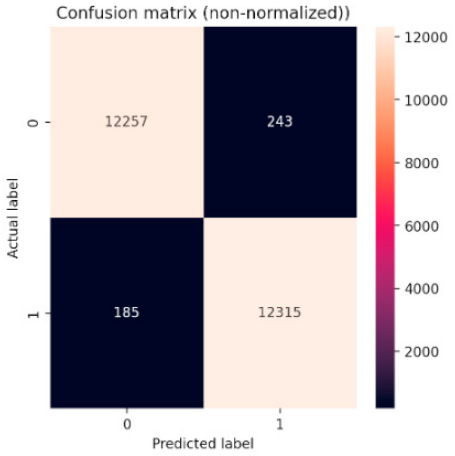


Fig. 3. Confusion Matrix of BiGRU with glove embedding and attention

competitive accuracy of 91.8%. Another study utilized Bidirectional Gated Recurrent Unit (BiGRU) with fine-grained attention, achieving an accuracy of 90.9%. The combination of Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) resulted in an accuracy of 88.9%. In contrast, our proposed method, employing Bidirectional Gated Recurrent Unit (BiGRU) with GloVe embedding and attention mechanisms, outperformed these methods with an impressive accuracy of 98%. The results highlight the effectiveness of our proposed model in achieving superior sentiment analysis performance on IMDb movie review datasets. Table 4 compares the accuracy of different sentiment

Author	**Models**	**Accuracy(%)**
Zhilin Yang et al.[10]	XLNet	96.21
Gang Liu et al.[11]	BiLSTM with attention and CNN	91.8
Xuanzhen Feng et al.[9]	BiGRU and fine grained attention	90.9
Camacho-Collados et al.[12]	CNN+LSTM	88.9
Proposed Method	**BiGRU + GloVe Embedding + Attention**	**98**

Table 4. Performance Analysis of the Proposed Method

analysis models. Zhilin Yang et al. achieved an accuracy of 96.21% using XLNet. Gang Liu et al. employed a com-

bination of BiLSTM with attention and CNN, achieving an accuracy of 91.8%. Xuanzhen Feng et al. utilized BiGRU with fine-grained attention, resulting in an accuracy of 90.9%. Camacho-Collados et al. adopted a CNN+LSTM approach, achieving an accuracy of 88.9%. The proposed method, which combines BiGRU with GloVe Embedding and Attention, outperforms the others with an accuracy of 98%. This suggests that the proposed method exhibits superior performance in sentiment analysis, surpassing the accuracy achieved by the other referenced models.

5. Conclusion

This work has successfully demonstrated the effectiveness of leveraging GloVe embeddings, Bi-GRU, and attention mechanism for movie review classification tasks. The combination of these advanced techniques has resulted in a remarkable accuracy of 98%, showcasing the potential of the proposed approach in tackling real-world challenges in natural language processing. The utilization of GloVe embeddings has proven to be a key factor in capturing semantic relationships and contextual information within the review text. These pre-trained word embeddings have enabled the model to learn meaningful representations of words and better comprehend the underlying structure of the input data. The incorporation of Bi-GRU has brought bidirectionality to the model, allowing it to capture both past and future contexts of each word in the input sequence. This feature has greatly improved the ability of model to understand the sequential nature of textual data and effectively capture long-range dependencies.The work could potentially be extended by providing the movie review summary of positive classification and negative classification. The proposal for the future work is classifying the reviews of a movie into positive summary and negative summary. This help the users to understand about the general opinion about the movie from the summary and audience can decide whether to watch the movie or not. The ability of model to generalize depends on the diversity of movie reviews in the training set, making it sensitive to biases or limitations in the data. If the training data lacks inclusiveness, the model may not produce better results.

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